Historical Market Matching with Persistent Homology

An Empirical Study of Component and Loop Information

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Abstract

Does adding persistent loops change historical market matches or their subsequent-return estimates? We examine this question by removing H_1 from a fixed retrieval procedure based on rolling cross-asset correlations. In a frozen eight-asset sample with 653 aligned price observations, the combined $H_0 + H_1$ method and H_0 alone return identical ordered references at all 23 attempted origins and identical predictions on all seven scored origins. In an overlapping historical extension, reference sets agree at all 32 attempts, with one ordering change on an abstained origin; all nine scored predictions remain identical. These results occur despite nonempty H_1 in 20.0% and 16.4% of the respective rolling-window samples. Against labelled correlation geometry, combined topology has slightly higher mean absolute error on the five and six shared scored dates, with coverage below one third. The small paired samples preclude a general forecasting comparison. We explain the representation's loss of asset identity and the conditions under which empty loop diagrams preserve retrieval order. Accompanying code and derived records support checks of every reported scoring denominator, returned reference list, paired error, and selected persistence fixture. An interactive website supports inspection and replay. The findings show why the presence of topological features and numerical correctness should be evaluated separately from their contribution to a prediction task.

Keywords: persistent homology; financial networks; historical analogues; selective prediction; walk-forward evaluation; reproducibility.

1 Introduction

Historical market matching asks which earlier periods resemble a current observation window and whether their subsequent outcomes offer useful reference information. The resemblance may concern a price path, a return distribution, or the dependence structure of a group of assets. This paper studies the last interpretation. Its empirical question is whether a topological summary of cross-asset relationships selects useful historical references for estimating a fixed target asset's subsequent return.

Cross-asset dependence affects diversification and aggregate risk because it describes how assets move together [1]. Correlation-based representations have long been used to extract financial hierarchies, track changing market structure, and filter networks of relationships [2–5]. Research on regime variation likewise motivates studying dependence as an evolving object [6, 7]. Whether similar dependence structures are followed by similar target returns is the empirical question considered here.

Persistent homology provides a multiscale description of connected components and loops. Rather than choosing one graph threshold, it records when features appear and disappear as the threshold changes. Its application to data analysis is well established [8, 9], and financial applications have studied critical transitions and possible early-warning patterns [10–13]. Here, persistence summarizes the distances among assets within a window. Those summaries rank earlier windows, whose later returns supply a target-return estimate.

Persistent homology compresses estimated correlations into a smaller description of dependence geometry. That compression may make common patterns easier to compare, but it also discards asset identities that may matter for the target return. The evaluation follows this tradeoff from the representation through reference selection to prediction error.

Three research questions organize the analysis. First, how does topological compression change the comparison of market windows relative to a labelled representation of the same geometry? Second, do loop features (H_1) change reference selection or prediction relative to connected-component features (H_0)? Third, how much of the evaluation sample survives the abstention rule, which withholds an estimate when references fail the stability checks? Selection and conditional error must be reported together: error on the dates for which a method returns a result is not an unconditional measure of usefulness [14, 15].

The central finding is that adding loops leaves every scored prediction unchanged in both archived samples. This is a result of a controlled H_1 ablation within one fixed implementation. It is supported by reference-list comparisons on all 23 frozen and 32 extended attempted origins, including abstentions. The labelled comparator supplies a separate, smaller comparison of forecast errors: five shared dates in the frozen sample and six in the extension. Neither comparison shows a lower mean error for combined topology. Because the historical intervals overlap and the settings were inspected during development, these findings describe the archived runs rather than a general ranking of forecasting methods.

The study contributes a documented negative result with an explicit scope. First, it separates the occurrence of loop features from changes in selected references and predictions. Second, it relates the observed ablation to elementary properties of persistence: loss of asset labels, the minimum-spanning-tree interpretation of H_0, and empty-loop rescaling. Third, it reports acceptance decisions and supplies derived records and checkers so that the numerical and empirical claims can be examined separately. These distinctions matter when a richer representation is proposed for a target-specific prediction task: a feature can be present and correctly calculated while adding no change to the returned prediction.

I built SecondOrder Homology[1] to inspect the geometry, selected historical windows, and their outcomes in one workspace. The website implements the procedure and displays its evidence; the research question is the incremental contribution of the representation. Its main views are described after the empirical results.

The archived weights, basket, and gates remain unchanged after this comparison. Keeping these choices fixed limits further tuning after inspection of the historical results [16–18]. A separate, timestamped prospective protocol specifies a narrower test of the incremental role of H_1. Its outcomes remain pending.

[1] Research website: https://homology.secondorder.tools/?lang=en.

2 Related work

2.1 Financial dependence and filtered representations

The conversion of correlations into asset distances connects this study to financial network analysis. Minimum spanning trees and related hierarchical constructions recover economically interpretable organization from stock-return relationships [2, 4]. Their evolution can reflect changing market correlations [3]. Filtered graph constructions retain selected links and higher connectivity while reducing the complexity of a dense network [5]. Persistent homology provides another reduction, organized across distance thresholds rather than through a single filtered graph.

Empirical dependence estimates also contain substantial sampling noise. Random-matrix analyses demonstrate why observed financial correlation structure should not be interpreted as entirely informative [19, 20]. Covariance shrinkage provides one approach to regularization [21], while dynamic conditional correlation models explicitly represent time-varying dependence [22]. The archived topology pipeline omits identity shrinkage, which would mix the estimated correlation matrix with the identity matrix. Its results therefore describe the unshrunk estimator used during development.

Regime models and historical matching also answer different statistical questions. A regime-switching model assigns probabilistic structure to latent states and their dynamics [6]; financial allocation applications examine implications of such states for decisions [7]. The retrieval procedure studied here ranks earlier windows directly, without fitting a latent-state transition model.

2.2 Persistent homology and financial applications

Foundational persistence work formalizes the lifetimes of topological features and their computation [23–25]. Stability results relate perturbations of specified inputs to changes in persistence representations, under their respective assumptions [26–28]. Statistical methods address uncertainty in estimated topological features [29], while measure-based geometry provides further tools for robust inference [30]. These strands establish useful mathematical context but require separate assumptions when applied to temporally dependent financial observations.

Financial studies have used persistent homology to examine crash-related patterns, critical transitions in financial networks, cryptocurrency dynamics, and critical-slowing-down indicators [10–13]. Their data objects and endpoints vary. A point cloud whose points are time observations differs from a finite space whose points are assets and whose distances come from correlations over a window. A warning indicator also differs from a nearest-reference return estimate. Our evaluation uses assets as points and a subsequent target return as its endpoint.

Persistence landscapes and persistence images transform diagrams into functional or vector summaries [31, 32]. Surveys situate these representations within broader statistical and machine-learning workflows [9, 33]. We compare diagrams directly. Landscape- or image-based retrieval would be a distinct representation experiment.

FinTDA provides an existing software approach to computing persistence diagrams for financial market change [34]. Our study links dependence geometry to temporally eligible analogues, acceptance decisions, and same-date prediction errors, with a direct ablation of loop information.

2.3 Retrieval, abstention, and evaluation

Nearest-neighbor methods provide the conceptual connection between similarity and reference-based prediction [35, 36]. However, fixed reference counts, temporal dependence, reference separation, and selection gates distinguish our procedure from settings in which classical consistency results apply. Case-based financial prediction also uses similar pricing histories to select reference outcomes [36]. Our retrieval object is the dependence among assets within a window. The labelled-distance comparator holds that object fixed while testing what is lost by topological compression.

Selective classification emphasizes the relationship between conditional error and the fraction of observations on which a method returns a decision [14, 15]. We use this relationship as an evaluation principle for return estimation. The financial regression setting calls for its own assessment of the acceptance rule.

Dependent observations motivate block resampling [37, 38], and forecast evaluation motivates paired loss comparisons [39]. Repeated model search creates additional interpretive problems [16–18, 40, 41]. Cross-validation results also depend on the data-generating and dependence assumptions [42, 43]. The evaluation uses temporal eligibility, paired losses, and explicit abstention counts. The small historical paired samples are reported descriptively.

3 Data and study scope

3.1 Frozen and extended historical samples

The frozen development experiment contains eight instruments: AAPL, AMZN, META, MSFT, NVDA, QQQ, SPY, and TSLA. The prediction target is META throughout the historical experiments. The fixed basket includes individual stocks and broad equity exchange-traded funds (ETFs). It was selected during development and does not represent a point-in-time universe of all tradable assets.

The historical source is Marketstack. The archived application labels its input as provider-adjusted OHLCV with `adjustment: all`. The frozen sample has 653 aligned provider-adjusted open, high, low, close, and volume (OHLCV) observations per instrument, from October 13, 2023, through October 12, 2026. The original experiment is identified as release 0.21.0 and was frozen on October 13, 2026. A later acquisition requested five years and produced an extended sample with 832 aligned observations from October 18, 2022, through October 13, 2026. The usable common-date interval is shorter than the requested five years and overlaps most of the frozen sample.

Table 1: Historical sample definitions. Dates describe the retained common-date input. Both samples use the same eight instruments and META as the target.

Sample	Prices	Returns	First retained date	Last retained date
Frozen	653	652	2023-10-13	2026-10-12
Extended	832	831	2022-10-18	2026-10-13

3.2 Alignment, exclusions, and the observation clock

An input date is retained only when all instruments have valid, aligned adjusted OHLCV records. A valid bar has a valid calendar date, finite fields, positive prices, nonnegative volume, low $\leq$

$\min(\text{open}, \text{close})$, and high $\geq \max(\text{open}, \text{close})$. Records must have strictly increasing, unique dates. Dates with missing or rejected adjusted bars are excluded, without forward-filling or raw-price substitution. A reconstruction of the frozen interval using the later provider response reconciles 653 retained dates, 90 adjusted-only rejection dates, and ten additional raw-invalid or absent dates. The supplementary data audit includes the original per-instrument invalid-row and alignment-loss counts. The rejected provider rows and their complete date-level failure log were not preserved; the later reconstruction therefore describes the newer response, while the original exclusions remain documented only in aggregate.

A row surviving common-date alignment is a *retained observation*. Thirty retained prices produce 29 returns between adjacent retained dates. If a session is absent, such a return can span multiple trading sessions. Likewise, a 20-observation target return uses the twentieth subsequent retained close; it is not necessarily a return over 20 consecutive exchange sessions. Across the 23 frozen attempted origins, these outcome intervals span 28–49 calendar days. Exchange closures, missing sessions, and quality exclusions are separate categories in the date audit.

Local resampling uses the same ordered sequence of retained-step returns. Calendar durations can therefore differ across both original and resampled windows.

3.3 Provenance and revision limitations

Marketstack describes adjusted prices as accounting for corporate actions such as splits and dividends, following its stated CRSP-based convention.[2] The study uses the supplied adjusted fields. Corporate-action factors and reinvestment are not independently reconstructed. Candle consistency checks assess the supplied rows, while point-in-time accuracy also depends on the history of provider revisions. In the later reconstruction, all frozen retained dates remain, but three retained asset-date bars differ from the frozen response. The original sample is preserved for evaluation.

The analysis therefore combines a past-only retrieval rule with a retrospective data source. Past-only chronology prevents explicit use of later window outcomes in the retrieval calculation. It does not by itself remove survivorship, instrument-selection, or revised-history effects. Dataset identities, source conventions, exclusions, parameters, and calculation provenance are necessary parts of reproducibility [44].

4 Mathematical framework

4.1 Window-local dependence geometry

Let p denote the number of instruments, w the number of retained prices in a query window, and $m = w - 1$ its number of retained-step returns. Instrument indices i, j range from 1 to p; k indexes retained dates, and t is the query endpoint. Write $P_{i,k} > 0$ for the adjusted close of instrument i at retained index k. For a window ending at t, the returns are

$$r_{i,k} = \log P_{i,k} - \log P_{i,k-1}, \qquad k = t - w + 2, \ldots, t. \tag{1}$$

All logarithms are natural. All instruments use the same retained dates; correlations use the complete common-date window.

[2] Provider documentation: https://docs.apilayer.com/marketstack/docs/api-endpoints-v1.

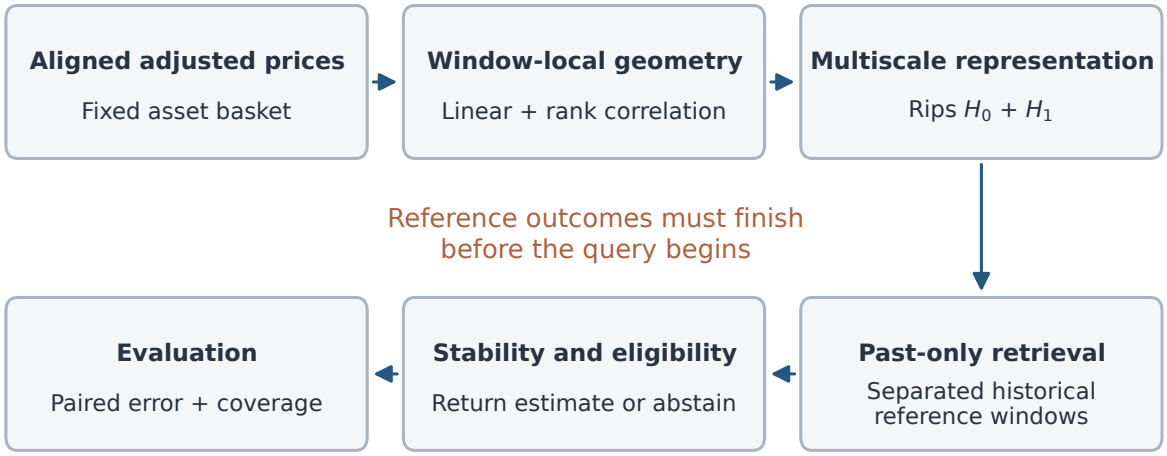


Figure 1: The implemented analysis chain. Geometry and persistent homology are calculated within each window. Historical references are eligible only after their complete outcomes precede the query start. Error and coverage are evaluated together.

For the linear channel, define $a_i = \mathrm{median}_k\, r_{i,k}$ and a robust scale

$$s_i = \max\left\{10^{-6},\, 1.482602218505602\ \mathrm{median}_k\,|r_{i,k} - a_i|\right\}. \tag{2}$$

The returns used in that channel are clipped to $[a_i - 4s_i, a_i + 4s_i]$ and then centered by their arithmetic mean and standardized by their population standard deviation within the window. Let the resulting sequence be $z^{(L)}_{i,k}$. For the rank channel, average ranks are assigned to ties in the original, unclipped returns; the ranks are then mean-centered and standardized, producing $z^{(R)}_{i,k}$. Correlations are

$$C^{(c)}_{ij,t} = \frac{1}{m}\sum_k z^{(c)}_{i,k} z^{(c)}_{j,k}, \qquad c \in \{L, R\}. \tag{3}$$

The calculation is unavailable if an instrument has insufficient variation. The implemented linear standard-deviation floor for this check is 10^{-12}; the rank channel requires positive variation.

The topology pipeline uses the empirical Gram matrices without identity shrinkage. Short-window estimation uncertainty remains relevant; covariance regularization would define an alternative estimator [21].

Each channel is converted into a chordal distance matrix:

$$d^{(c)}_{ij,t} = \sqrt{2(1 - C^{(c)}_{ij,t})}. \tag{4}$$

For standardized common-date vectors, $v^{(c)}_i = z^{(c)}_i/\sqrt{m}$ has unit norm and $d^{(c)}_{ij,t} = \|v^{(c)}_i - v^{(c)}_j\|_2$. Here $\|x\|_2 = (\sum_j x_j^2)^{1/2}$ is the Euclidean norm. This realization allows coincident points in degenerate cases. An arbitrary matrix assembled from correlations estimated on different subsets need not have this property. Floating-point clipping of correlations to $[-1, 1]$ and explicit numerical tolerances are implementation safeguards.

4.2 Rips persistence and diagram comparison

For a $p \times p$ distance matrix D, a simplex σ is a nonempty subset of the asset vertices $\{1, \dots, p\}$. The Vietoris–Rips complex at threshold $\epsilon \geq 0$ is

$$\mathrm{VR}_\epsilon(D) = \{\sigma : \ d_{ij} \leq \epsilon \text{ for all } i, j \in \sigma\}. \tag{5}$$

A simplex with $q+1$ vertices has dimension q: vertices, edges, and filled triangles have dimensions zero, one, and two. As ϵ increases, these nested complexes form a filtration. Edges enter at their distances, triangles at their largest edge distance, and all vertices at zero.

For a complex K, let $C_q(K; \mathbb{F}_2)$ be the vector space generated by its q-dimensional simplices over the field $\mathbb{F}_2 = \{0, 1\}$, with arithmetic modulo two. The boundary map ∂_q sends a simplex to the sum of its codimension-one faces, and $\partial_0 = 0$. Since $\partial_q \partial_{q+1} = 0$, boundaries are cycles. Its homology is

$$H_q(K; \mathbb{F}_2) = \ker \partial_q / \operatorname{im} \partial_{q+1}. \tag{6}$$

Thus H_0 describes connected components and H_1 describes cycles modulo boundaries of filled triangles. The Betti number $\beta_q(\epsilon) = \dim_{\mathbb{F}_2} H_q(\mathrm{VR}_\epsilon(D); \mathbb{F}_2)$ counts the corresponding independent features at one threshold. Vertices, edges, and triangles, collectively the two-skeleton, suffice to calculate these two dimensions.

Persistence tracks features through the filtration. A diagram point (b, d) records the birth threshold b and death threshold d of a class; its lifetime is $d - b$. The diagram is a multiset because several classes can have the same coordinates. A class that never dies is essential. We calculate persistence by finite boundary-matrix reduction [23, 24]. Equal filtration values are ordered by dimension and then vertex indices. Zero-lifetime classes and the one essential H_0 class are omitted. See Edelsbrunner and Harer [25], Bauer [45] for computational background.

Let $\mathcal{P}_t^{(c,q)}$ be the resulting finite diagram for channel $c \in \{L, R\}$ and dimension $q \in \{0, 1\}$. For finite diagrams A, B, let M range over partial one-to-one matchings of their point instances. Write A_M, B_M for the matched instances. The order-two diagram Wasserstein distance with L^∞ ground norm is

$$\begin{aligned} W_2(A, B)^2 = \min_M \Bigg\{ & \sum_{(x,y) \in M} \|x - y\|_\infty^2 \\ & + \sum_{(b,d) \in A \setminus A_M} \frac{(d-b)^2}{4} + \sum_{(b,d) \in B \setminus B_M} \frac{(d-b)^2}{4} \Bigg\}. \end{aligned} \tag{7}$$

Here $\|x\|_\infty = \max_j |x_j|$. Each unmatched point is sent to the diagonal $\{(u, u) : u \in \mathbb{R}\}$, at squared cost $(d-b)^2/4$; W_2 is the nonnegative square root. This finite formulation is equivalent to adding diagonal copies and solving an assignment problem. The implementation uses that assignment formulation without entropic approximation [46, 47].

For two windows ending at t and s, define

$$u_{c,q}(t, s) = \frac{W_2(\mathcal{P}_t^{(c,q)}, \mathcal{P}_s^{(c,q)})}{\sqrt{p}}. \tag{8}$$

The full distance and its dimension-specific counterparts are

$$D_{\text{full}}(t,s)^2 = \sum_{c\in\{L,R\}} \alpha_c \left[\tfrac{1}{2}u_{c,0}(t,s)^2 + \tfrac{1}{2}u_{c,1}(t,s)^2\right], \tag{9}$$

$$D_q(t,s)^2 = \sum_c \alpha_c u_{c,q}(t,s)^2, \qquad q = 0,1, \tag{10}$$

with $(\alpha_L, \alpha_R) = (0.7, 0.3)$. These weights were inherited from the development implementation and are held fixed.

The labelled comparator uses the same channel-specific distance matrices, retaining asset identities:

$$L_c(t,s)^2 = \frac{1}{\binom{p}{2}} \sum_{i<j} \left(d^{(c)}_{ij,t} - d^{(c)}_{ij,s}\right)^2, \tag{11}$$

$$D_{\text{lab}}(t,s)^2 = 0.7L_L(t,s)^2 + 0.3L_R(t,s)^2. \tag{12}$$

Both procedures use the same unshrunk geometry; their difference is the information retained when comparing windows.

4.3 What the representation preserves

The following elementary properties describe the compression produced by persistence [25, 28].

Proposition 1 (Asset-label invariance)**.** *If Π is a permutation matrix, D and $\Pi D \Pi^{\mathsf{T}}$ have the same finite Rips persistence diagrams in dimensions zero and one. A labelled comparison with a fixed second window can change under this permutation.*

Proposition 2 (The finite H_0 summary)**.** *For a complete finite weighted graph induced by D, the positive finite H_0 death values are the positive merge-edge weights of a minimum spanning tree, counted with multiplicity. The finite diagram omits which labelled edges produced those merges.*

Proposition 3 (Empty-H_1 rescaling)**.** *If the query and every eligible candidate have empty H_1 diagrams in both channels, then $D_{\text{full}}(t,s) = D_0(t,s)/\sqrt{2}$ for each candidate. With identical deterministic tie-breaking and separation rules, the returned reference ordering is unchanged before the acceptance gates are applied.*

Proofs appear in Appendix A. Proposition 3 applies to the unperturbed query and candidate bank. If a resampled query has nonempty H_1, the two methods' stability diagnostics need not agree. The empirical audit checks the returned references and predictions directly.

Figure 2 gives an analytic example. Four unit vectors at the cardinal directions of a plane have adjacent distance $\sqrt{2}$ and opposite distance 2. Their H_0 diagram has three points $(0, \sqrt{2})$ and their H_1 diagram has one point $(\sqrt{2}, 2)$. Reassigning asset labels preserves both diagrams while changing some labelled edges. The construction illustrates how identical dependence shapes can assign different roles to named instruments.

4.4 Stability is a sequence of separate questions

The stability of persistence diagrams under specified perturbations is a mathematical property with explicit assumptions [26–28]. Statistical confidence sets for diagrams require further modeling

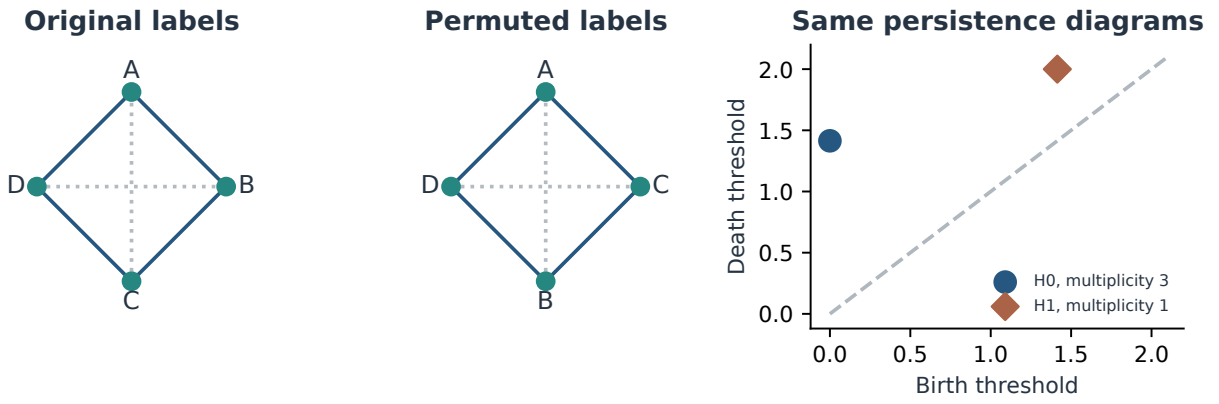


Figure 2: Analytical four-asset example. Solid edges have distance $\sqrt{2}$; dotted diagonals have distance 2. The two label assignments produce the same H_0 and H_1 diagrams, although a labelled distance comparison can distinguish them. The diagram excludes the essential component.

conditions [29]. Neither directly implies that the nearest historical window is unchanged. Even a small distance perturbation may exchange two candidates with nearly equal distances. Conversely, changed distances may leave the selected references unchanged when margins are large.

Prediction introduces another step. Stable references may have heterogeneous subsequent returns, and the most relevant structural features for the selected target may be precisely those discarded by the topological summary. We therefore distinguish uncertainty in estimated correlations, diagram perturbation, reference-order perturbation, and target-return error. The local resampling rule in the next section addresses only a restricted reference-sensitivity question.

5 Historical retrieval and selective prediction

5.1 Chronology and separated references

The historical evaluation fixes $w = 30$ retained prices and horizon $h = 20$ retained steps. A query ending at t begins at $a_t = t - w + 1$. Candidate reference endpoints are examined every five retained observations, beginning with the first complete window. A candidate ending at s is eligible only when its full reference window is valid and its target outcome is available strictly before the query starts:

$$s + h < a_t. \tag{13}$$

This rule is stricter than merely requiring the reference window to end before the query endpoint. It excludes outcomes that extend into the query period.

Eligible candidates are sorted by the chosen distance, with earlier endpoint index used to break exact ties. References are accepted greedily from this ordering, subject to endpoint separation

$$|s^{(i)} - s^{(j)}| \geq w + h = 50 \tag{14}$$

for every pair of selected reference endpoints $s^{(i)}, s^{(j)}$. The rule selects up to six references and requires at least three for scoring. It limits reuse within each reference set; different queries can still share observations and historical information.

All compared methods share the source geometry, candidate chronology, candidate stride, maximum reference count, separation rule, and downstream target-return calculation. Their selected references and stability diagnostics can differ because their distances differ. The comparison evaluates each fixed retrieval procedure together with its selection behavior.

5.2 Return estimate and historical baselines

The realized target return, in percent, is

$$Y_t^{(h)} = 100 \left(\frac{P_{\mathrm{META},t+h}}{P_{\mathrm{META},t}} - 1 \right). \tag{15}$$

The point estimate is the unweighted arithmetic mean of the selected references' mature target returns:

$$\widehat{Y}_{\mu,t} = \frac{1}{|\mathcal{N}_{\mu,t}|} \sum_{s \in \mathcal{N}_{\mu,t}} Y_s^{(h)}, \tag{16}$$

where $\mu \in \{\mathrm{full}, 0, 1, \mathrm{lab}\}$ identifies the retrieval method and $\mathcal{N}_{\mu,t}$ is its set of selected reference endpoints. The set size $|\mathcal{N}_{\mu,t}|$ can vary from three to six. Throughout the evaluation, Y_t abbreviates $Y_t^{(h)}$ for the fixed horizon $h = 20$. Distances determine which mature reference returns enter the unweighted mean.

The stored evaluation also includes a historical-mean baseline constructed from earlier reference outcomes with a fixed 50-observation endpoint stride. Those outcomes satisfy the same requirement of maturity before the query begins, and at least three are required. A zero-return estimate is included as a transparent arithmetic benchmark on shared dates. These baselines help describe the scale of error; the historical-mean bank is distinct from the nearest-reference bank.

The estimate is a local average in the sense of nearest-neighbor regression [35, 36]. Here the reference count remains fixed, and chronology, separation, and acceptance gates determine which observations enter that average.

5.3 Local block-resampling diagnostic

For each query, the implementation constructs eight joint circular-block resamples of its $m = 29$ retained-step return vectors. All instruments share the resampled indices, preserving the observed contemporaneous pairing. Block length is

$$\ell = \max\{2, \mathrm{round}(\sqrt{m})\} = 5. \tag{17}$$

Resampled returns are accumulated from each instrument's first query close to form an artificial price sequence. The diagnostic therefore perturbs the estimated query geometry. Historical candidate geometries and historical outcomes remain fixed.

The comparison bank for this diagnostic is local: the first 30 eligible candidates in the original ordering, augmented if necessary by any of the first three selected original references. For each valid resample, candidates in this shortlist are re-ranked and up to three separated references are selected. The first retention statistic is the fraction of valid resamples in which the original best reference remains anywhere in the selected top three. The second is the average fraction of the original three-reference set retained. Scoring requires both fractions to be at least 0.5 and at least four valid resamples.

This query-sensitivity calculation holds the historical shortlist fixed. Block resampling respects short-range dependence [37, 38]; these eight replicates and their thresholds are a diagnostic rather than a calibrated confidence procedure. In particular, deterministic tie-breaking can make some repeated selections appear stable even when their distances are uninformative.

5.4 Acceptance and abstention

Additional gates require a valid query geometry, a sufficiently long window, a usable mature outcome, enough separated references, and enough earlier baseline outcomes. A query is not scored if more than 20% of an instrument's retained-step returns are clipped. A bank whose eligible distances are all numerically indistinguishable also triggers abstention, using a relative tolerance of 10^{-12} with unit-scale protection. The full implementation preserves every failed check as well as one primary reason per abstained origin.

An origin is accepted, or scored, when all gates pass; otherwise the procedure abstains and returns no estimate. Errors are calculated on accepted origins, while coverage counts all attempted origins [14, 15]. An economic cost for abstaining would require a specified decision problem.

6 Evaluation and numerical verification

6.1 Walk-forward endpoints and paired comparisons

The historical validation begins at zero-based retained endpoint 180 and advances by 20 retained observations. A query enters the attempted-origin count once its outcome is mature. Consequently, the frozen and extended samples have 23 and 32 attempted origins. The final live analysis window in a dataset need not itself be a mature evaluation origin.

Let $A_{\mu,t} \in \{0, 1\}$ denote whether method μ scores attempted origin t. Its coverage is

$$\mathrm{Covg}_\mu = \frac{1}{N} \sum_{t \in \mathcal{T}} A_{\mu,t}, \tag{18}$$

where $\mathcal{T}$ is the attempted set and $N = |\mathcal{T}|$. Let $n_\mu = \sum_{t \in \mathcal{T}} A_{\mu,t}$ be the number of scored origins. The mean absolute error on that method's scored set is

$$\mathrm{MAE}_\mu = \frac{1}{n_\mu} \sum_{\substack{t \in \mathcal{T} \\ A_{\mu,t}=1}} |Y_t - \widehat{Y}_{\mu,t}|, \qquad n_\mu > 0. \tag{19}$$

Because accepted sets differ, these own-set MAEs describe performance on different dates.

For two methods μ, ν, define the shared scored set

$$S_{\mu,\nu} = \{t \in \mathcal{T} : A_{\mu,t} = A_{\nu,t} = 1\}. \tag{20}$$

For $|S_{\mu,\nu}| > 0$, the paired loss difference is

$$\Delta_{\mu,\nu} = \frac{1}{|S_{\mu,\nu}|} \sum_{t \in S_{\mu,\nu}} \left(|Y_t - \widehat{Y}_{\mu,t}| - |Y_t - \widehat{Y}_{\nu,t}| \right). \tag{21}$$

Negative values favor μ. Returns are expressed in percent and absolute errors and loss differences in percentage points (pp). We report both methods' coverage and the paired denominator alongside this conditional comparison. The paired difference is conditional on both methods accepting the date.

Paired forecast-loss comparison is a standard organizing principle [39]. The stored implementation produces uncertainty intervals only when at least 20 shared scored origins are available. It uses 2,000 paired circular-block percentile bootstrap replicates, seed 1101, and block lengths 1, 2, and 3; reported bounds are the empirical 2.5th and 97.5th percentiles. This is distinct from the eight query-sensitivity resamples. Neither historical topology–labelled comparison reaches the minimum paired count. We report the observed rows and averages without a significance test.

Block resampling across the ordered shared scored origins also has an interpretive limitation: consecutive elements of that selected sequence may be separated by different amounts of calendar or retained-observation time. Future uncertainty results must retain their paired-date denominator and discuss whether that dependence approximation is adequate.

6.2 Reference and feature diagnostics

To isolate the incremental role of H_1, we compare the ordered full-topology and H_0 reference lists at every attempted origin where both return a nonempty list. For nonempty endpoint sets U, V, their Jaccard similarity is $J(U,V) = |U \cap V|/|U \cup V|$. We report this value, changes in ordering, and changes in the best reference. On shared scored origins we additionally compare predictions, using absolute tolerance 10^{-12} for numerical equality. The audit covers returned lists; ranks of unselected candidates can still differ.

Feature occurrence is assessed separately over every valid trailing 30-price window. We count windows with a nonempty H_1 diagram in either channel and summarize positive lifetimes $d - b$. Rolling windows share observations, and the two channels use the same returns. Occurrence and pooled lifetimes are therefore reported descriptively.

6.3 Replay and independent arithmetic checks

Both historical datasets were rerun using the inspected retrieval implementation. The frozen topology–labelled paired count and loss difference reproduce as five and +0.4382884011544448 pp. The extended paired count and difference reproduce as six and +1.0425944582169537 pp. Coverage and the full/H_0 equality conclusions also agree with their archived records.

We supplemented replay with an independently written Python boundary-reduction calculation and SciPy linear-sum assignment. The fixture set contains 48 channel-specific distance matrices from 24 evenly spaced frozen windows, 50 distance matrices from generated unit-vector configurations with three to eight points, and the analytical four-point square. Across these 99 matrices, all 198 finite H_0/H_1 diagrams agree exactly in their exported coordinates. Forty diagram-distance fixtures agree within a maximum absolute difference of 5.56×10^{-17}, below the specified 10^{-12} tolerance. The selected fixtures and independent checker accompany the source submission in `anc/verification/`.

A separate arithmetic and chronology audit checks all four methods in both samples against the archived records. It reproduces scored counts and MAEs, the five- and six-date paired comparisons, and full/H_0 prediction equality. Across the returned lists of all four procedures, 459 frozen and 674 extended references satisfy the 30-price window, 50-observation separation, and outcome-maturity

conditions. The audit also detects the extension's one order change described in Section 7.6. These checks operate on derived records and retained-date lists. The supplementary export script regenerates the three empirical data tables, a loop-ablation summary, and CSV files containing all attempted origins, returned references, scored predictions, and paired losses. A file manifest records input and script hashes.

The independent numerical checks cover filtration reduction and diagram matching on the exported fixtures. The historical audit checks the supplied records; it does not recompute the complete retrieval and gating process from prices. A further implementation comparison could use Ripser or Ripser.py [45, 48].

6.4 Development history and multiplicity

The reported windows, target, basket, and inherited weights were inspected during development. The extension was acquired and analyzed after the frozen experiment. Both are development samples, with the associated model-selection concerns [16–18, 40, 41].

Temporal eligibility follows guidance on preserving dependence in validation [43]. Results for autoregressive cross-validation depend on the assumed process [42]. In this design, overlapping queries and shared historical banks remain sources of dependence.

7 Empirical results

7.1 What the loop ablation changes

Table 2 states the main result before examining an individual match or the labelled comparator. The full representation and H_0 alone select the same reference sets at every attempted origin in both samples. Ordered lists agree at all frozen origins and at 31 of 32 extended origins. The extension's only order change occurs after the first reference on an abstained date and changes neither the selected set nor a scored prediction. Scoring decisions agree for the two methods, and paired MAE differences are exactly zero on all seven and nine shared scored dates.

Table 2: Incremental effect of adding H_1 to the fixed retrieval procedure. Reference comparisons include abstained origins. Prediction comparisons include only shared scored origins; both methods accept the same origins. The samples overlap.

Comparison	Frozen	Extended
Identical returned reference sets	23/23	32/32
Identical ordered reference lists	23/23	31/32
Identical shared scored predictions	7/7	9/9
Full minus H_0 paired MAE (pp)	0.000000	0.000000
Windows with H_1 in either channel	125/624	132/803

Loop features are present in both samples, while the returned predictions are unchanged. The reference-list comparison covers all attempted origins, including those that fail an acceptance gate; the prediction comparison uses the seven and nine shared scored origins.

7.2 A complete historical example

The first shared scored origin, January 16, 2025, provides a concrete view of the full calculation. The chronological selection rule avoids choosing the example for favorable error. Its query contains 30 retained prices from November 26, 2024, through January 16, 2025. The realized 20-observation outcome ends on February 19, 2025, spanning 34 calendar days. Figure 3 displays the query's two dependence channels and their finite persistence diagrams.

There are 33 eligible candidates before the separation rule. Full topology returns three reference windows, in order: June 12–August 1, 2024; October 13–December 18, 2023; and February 1–March 28, 2024. Their combined distances are 0.03313, 0.04492, and 0.06167. Every reference's complete 20-observation outcome ends before November 26, 2024. Thus, the query's own later return has no role in selecting these references.

The reference lists are also unchanged when H_1 is removed. In the local eight-resample diagnostic, the original best reference is retained in the selected top three in four resamples, giving retention 0.50. Average retention of the original three-reference set is 0.5417. Both satisfy the fixed 0.50 gate, and the origin is scored.

Figure 4 aligns each reference at its endpoint and shows its subsequent META return path. These are target price paths associated with structurally selected windows; similarity of their price shapes was not the retrieval criterion. Their dispersion illustrates the gap between matching dependence and estimating the direction or magnitude of one instrument's return.

The three mature reference returns average 2.1573%, while META's realized query outcome is 15.1268%. The resulting absolute error is 12.9695 pp. Labelled geometry estimates 12.2623%, with absolute error 2.8645 pp. The paired loss difference is therefore +10.1049 pp, favoring the labelled comparator on this origin. The earlier historical-mean baseline estimates 0.7779%, with error 14.3488 pp.

The origin passes the sensitivity gate, yet its reference mean differs substantially from the realized return. Reference stability and return accuracy are therefore separate empirical questions. The aggregate evaluation retains every attempted date and the complete five-date paired comparison.

7.3 Coverage and same-date error

Table 3 reports every evaluated method, including the exploratory H_1-only ablation. In the frozen sample, full topology scores seven of 23 attempts, for 30.4% coverage. The labelled comparator scores 11, for 47.8% coverage. Full topology has 16 abstentions: 15 with unstable references as the primary reason and one with too few references. The labelled comparator has ten primarily unstable abstentions and two with too few references.

The full and labelled procedures share five scored frozen origins. Their respective MAEs on those dates are 15.672514 and 15.234225 pp, giving $\Delta_{\text{full,lab}} = +0.438288$ pp. The historical-mean and zero-return baselines have MAEs of 17.743222 and 18.498150 pp on the same five dates. Thus, the full method's error is lower than these two simple estimates in this selected comparison, but slightly higher than the labelled structural comparator.

Table 4 and Figure 7 show the variation behind the average: per-origin loss differences have both signs, and most attempted origins are excluded by at least one acceptance gate.

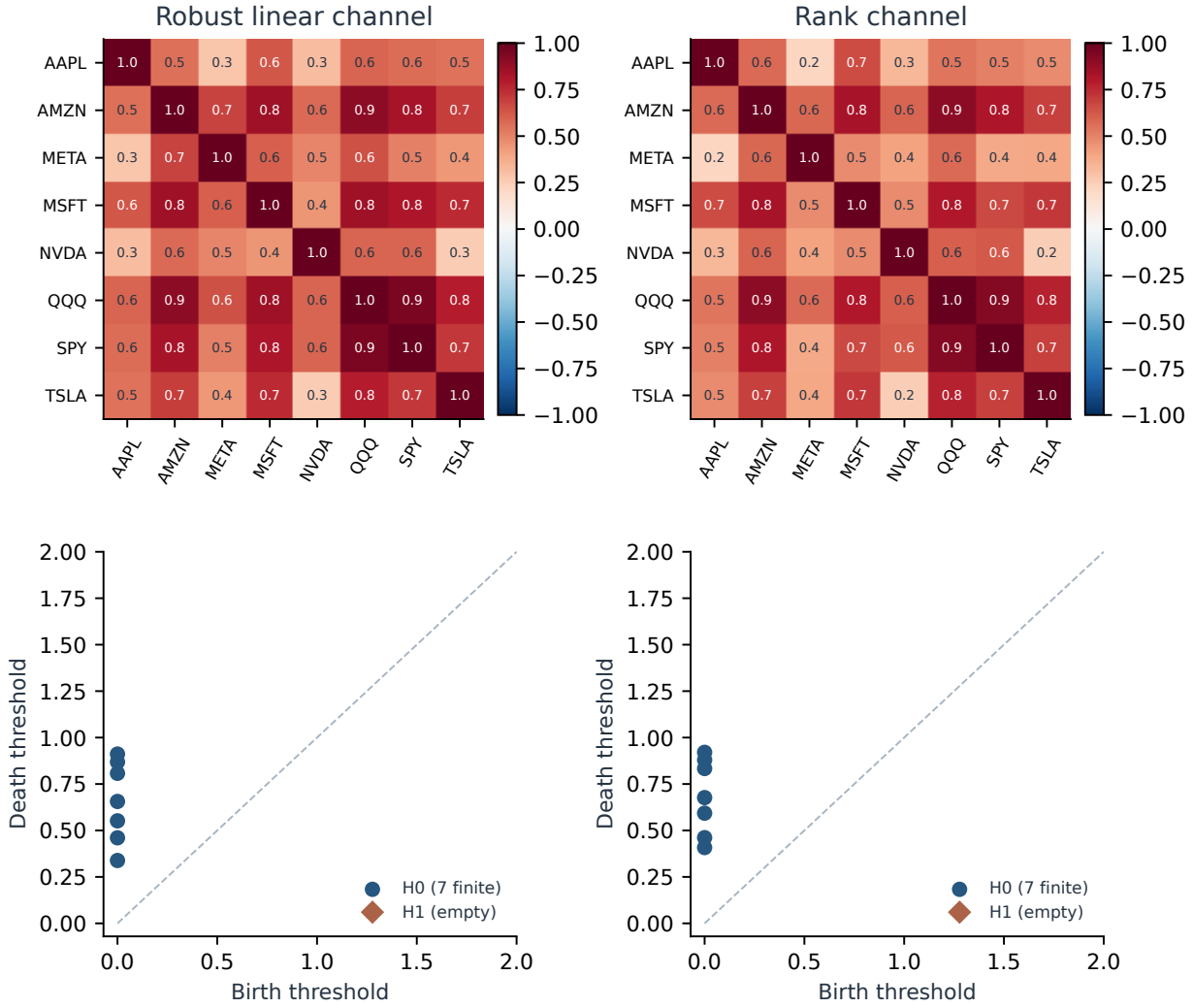


Figure 3: Query geometry for November 26, 2024–January 16, 2025. Top: robust linear and rank correlation matrices for the same eight assets. Bottom: corresponding finite persistence diagrams. Essential H_0 is omitted. The input is the frozen dataset.

Table 3: All attempted origins and method-specific scored sets. The final column uses each method's own scored dates; same-date comparisons appear in Table 4. H_1-only is an exploratory ablation.

Sample	Method	Attempts	Scored	Coverage (%)	Own-set MAE (pp)
Frozen	$H_0 + H_1$	23	7	30.4	13.998
Frozen	H_0 only	23	7	30.4	13.998
Frozen	H_1 only	23	22	95.7	10.139
Frozen	Labelled	23	11	47.8	9.759
Extended	$H_0 + H_1$	32	9	28.1	13.771
Extended	H_0 only	32	9	28.1	13.771
Extended	H_1 only	32	30	93.8	11.736
Extended	Labelled	32	16	50.0	13.640

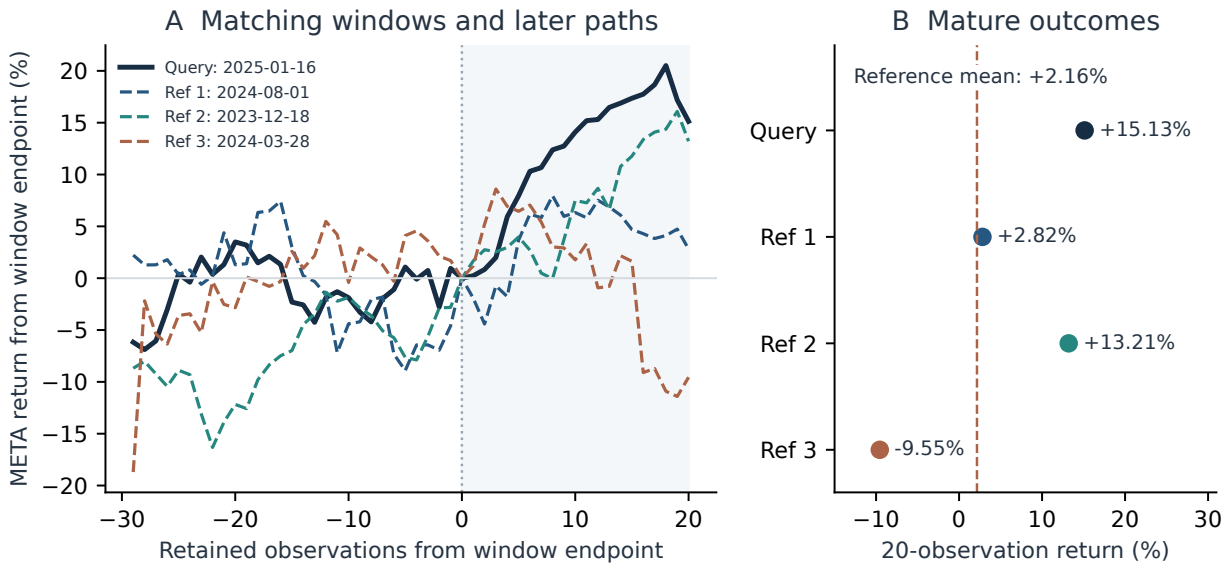


Figure 4: META paths for the first shared scored origin and its three topological references. Every path is rebased to zero at the end of its matching window. Negative indices show the 30-price matching window; positive indices show the 20-observation outcome. Reference outcomes were fully available before the query began. The query outcome is displayed retrospectively for evaluation.

Table 4: Complete frozen full-topology versus labelled comparison. Predictions and outcomes are META retained-20-observation returns in percent. The final column is the full minus labelled absolute-error difference in percentage points.

Query endpoint	Realized	Full prediction	Labelled prediction	Error difference
2025-01-16	15.127	2.157	12.262	+10.105
2025-04-16	27.497	10.934	11.064	+0.130
2025-07-17	11.508	9.584	1.636	-7.949
2025-08-14	-3.393	6.860	8.193	-1.333
2026-08-26	34.965	-1.687	-0.449	+1.238

7.4 Observed H_1 features and incremental retrieval

The frozen sample supplies 624 valid trailing 30-price windows. The linear channel has nonempty H_1 in 57 windows, the rank channel in 88, and at least one channel in 125 windows, or 20.0%. Across both channels, 158 positive finite-class instances have a median lifetime of 0.022584 correlation-distance units and a maximum of 0.138041. Figure 6 displays the occurrence counts and lifetime distribution.

For the frozen reference-set comparison in Table 2, mean Jaccard similarity is 1.000 and the best returned reference is unchanged at every attempted origin. Loop occurrence is measured across the full rolling history, so it describes a broader set of windows than those selected for prediction.

Several mechanisms are compatible with the result. A query and its relevant candidates may have weak or absent H_1; H_0 distance differences may dominate selection margins; or candidate changes may occur below the returned part of the ordering. The audit does not distinguish these mechanisms. Proposition 3 supplies a sufficient condition for ranking equality when the relevant diagrams are empty.

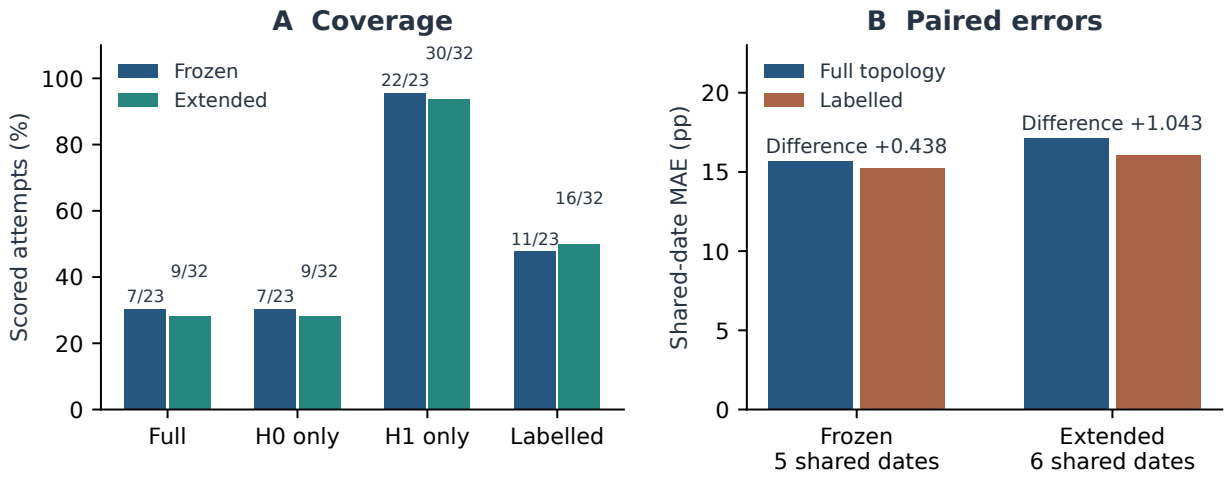


Figure 5: Coverage and shared-date prediction error. Panel A reports each method's scoring frequency, with numerator and denominator shown. Panel B compares full topology and labelled geometry only on their shared scored dates. The two samples cover overlapping history.

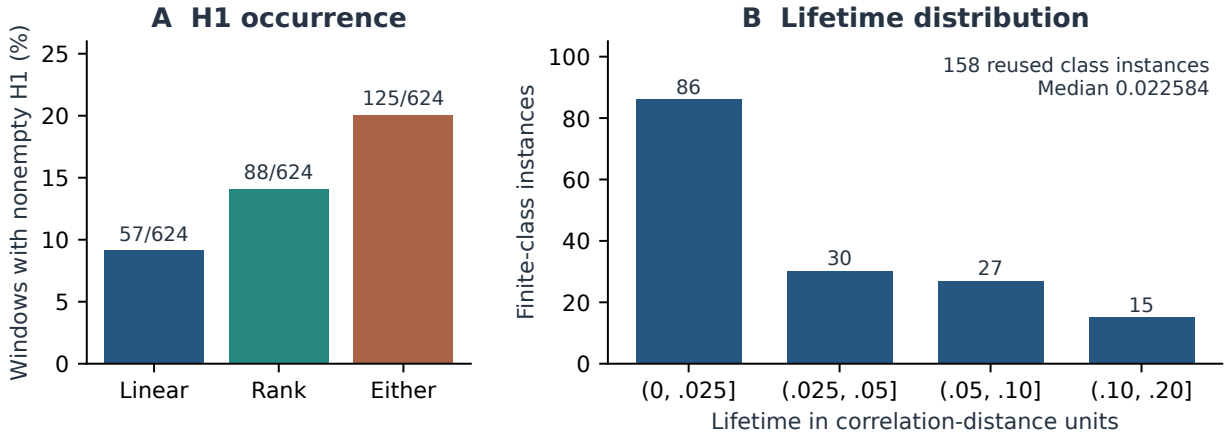


Figure 6: Frozen-sample H_1 diagnostics. Panel A counts windows, with a common denominator of 624. Panel B pools finite-class lifetimes across both channels and overlapping windows; its denominator is 158 reused class instances. All observed lifetimes lie below 0.2.

7.5 Exploratory H_1-only results

The H_1-only ablation scores 22 frozen origins and 30 extended origins, more than the full method. Its own-set errors in Table 3 therefore refer to substantially different dates. On the five dates shared by all four frozen methods, its MAE is 15.015164 pp, compared with 15.234225 pp for labelled geometry. On the six corresponding extended dates, the values are 14.404797 and 16.093999 pp. This exploratory comparison slightly favors H_1 alone over labelled geometry on those selected dates.

Interpreting this difference requires attention to ties. Sparse or empty diagrams can generate tied or nearly tied distances; the fixed ordering and separation rule can then select references for reasons only weakly related to distinct loop structure. Repeated selections under the same tie rule may also pass the local retention gate. The existing all-distances-indistinguishable check excludes some degenerate banks but does not resolve every partial tie. An H_1-only evaluation would need to isolate

the effect of tie-breaking on reference selection and use new outcomes. The locked prospective comparison instead tests the addition of H_1 to H_0.

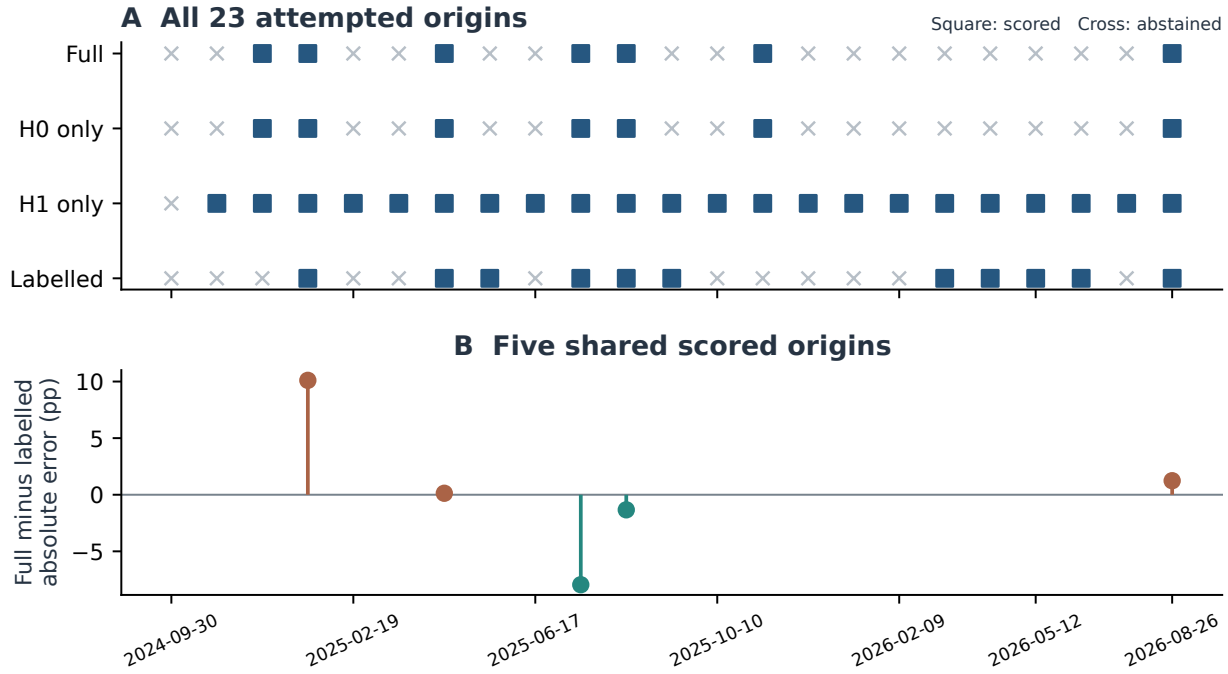


Figure 7: Frozen attempted-origin audit. Panel A shows all 23 origins for all four procedures. Panel B places the five paired full-minus-labelled loss differences at their corresponding origins; positive values favor labelled geometry. Identical full/H_0 acceptance patterns accompany identical returned reference lists.

7.6 Overlapping historical extension

In the extended sample, full topology scores nine of 32 attempts, or 28.1%, while labelled geometry scores 16, or 50.0%. The full method has 22 primarily unstable abstentions and one with too few references. The methods share six scored dates; the full and labelled MAEs are 17.136593 and 16.093999 pp, respectively, for a difference of +1.042594 pp. On these dates the stored historical-mean and zero-return errors are 12.104554 and 9.858429 pp. Thus, the favorable relation to those simple baselines in the frozen selected set does not persist in the extended selected set.

Full topology and H_0 return the same reference sets at all 32 attempted origins. Their ordering agrees at 31 origins. At the abstained origin of January 21, 2025, the third and fourth references exchange positions; the best reference is unchanged. All nine shared scored predictions are identical. The extended input provides 803 valid rolling windows, with H_1 in either channel in 132, or 16.4%. These are additional descriptive results under the inherited design.

The extension overlaps the frozen history, uses a later provider response with some revisions, and shifts the retained-index evaluation grid by adding earlier observations. Query dates, candidate banks, and selected subsets change together, so the comparisons remain separate.

8 Implementation and evidence inspection

8.1 Structural views

SecondOrder Homology is a bilingual research website linking an asset window, its dependence geometry, historical references, and archived evaluation. The target controls the displayed price path and return; the full basket defines the geometry. Historical results always use the archived adjusted samples.

Figure 8 shows the linked network and persistence diagram. Channel and dimension controls select linear or rank dependence and H_0 or H_1; a slider varies the filtration threshold. The circular network layout keeps asset names readable and does not encode distances. The diagram records finite birth–death pairs, and a barcode displays their intervals $[b, d)$. For diagram $\mathcal{P}$, the first landscape is $\lambda_1(x) = \max_{(b,d)\in\mathcal{P}} \max\{0, \min(x-b, d-x)\}$ [31]. For positive lifetimes $l_j = d_j - b_j$, the descriptive energy is $\sum_j l_j^2$, the longest lifetime is $\max_j l_j$, and entropy is $-\sum_j \pi_j \log \pi_j$, where $\pi_j = l_j / \sum_k l_k$. These summaries and the landscape are zero for empty diagrams. They are display summaries, not additional retrieval inputs.

8.2 Archived results and replay

The browser interface uses modular JavaScript calculation code, worker dispatch, and a server-side market-data endpoint. Each experiment records its dataset fingerprint, query, parameters, calculation identity, and results. Replay compares the recomputed record with the archived one; the frozen replay agreed on the input fingerprint, returned references, predictions, and errors [44]. The published interface version is 1.2; the original frozen calculation remains release 0.21.0. A later interface or provider refresh does not change the identities used in the paper.

The results page presents same-date error, method coverage, every attempted origin, and the H_1 ablation (Figure 9). The exploratory interface can show up to eight analogues at several horizons. The evaluated procedure instead uses the fixed three-to-six-reference rule and 20-retained-observation horizon. The supplementary Python scripts reproduce the data tables and expose the reference lists independently of the interface.

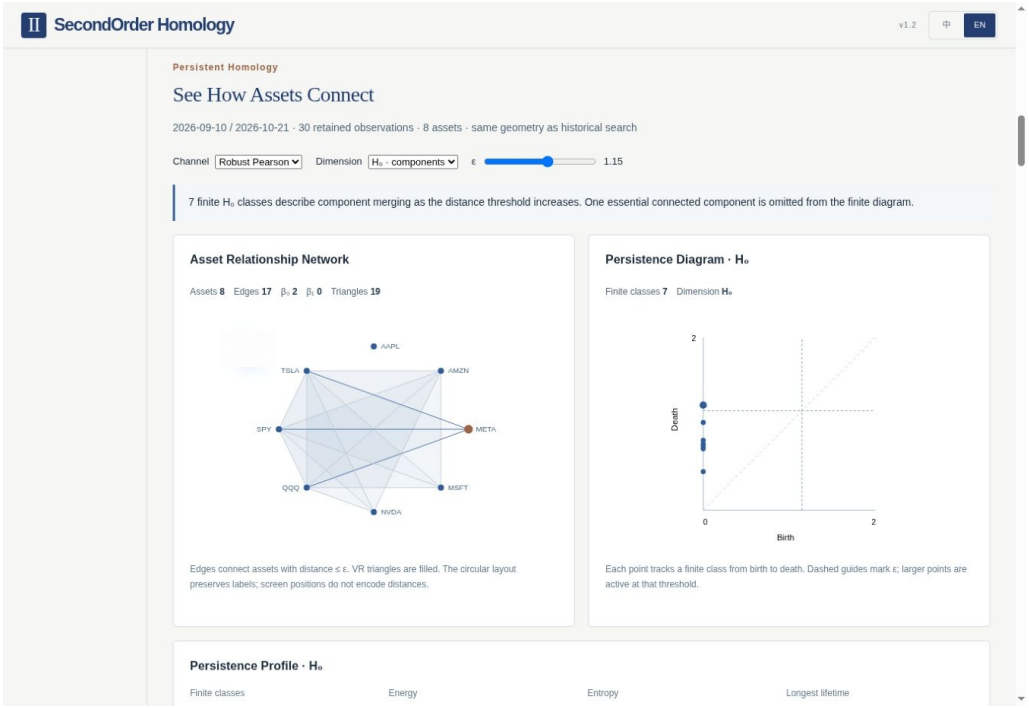


Figure 8: Linked network and persistence views in interface version 1.2. The selected provider-data window is September 10–October 21, 2026. The screenshot documents the implementation and supplies no additional forecast evidence. Network positions use a circular layout.

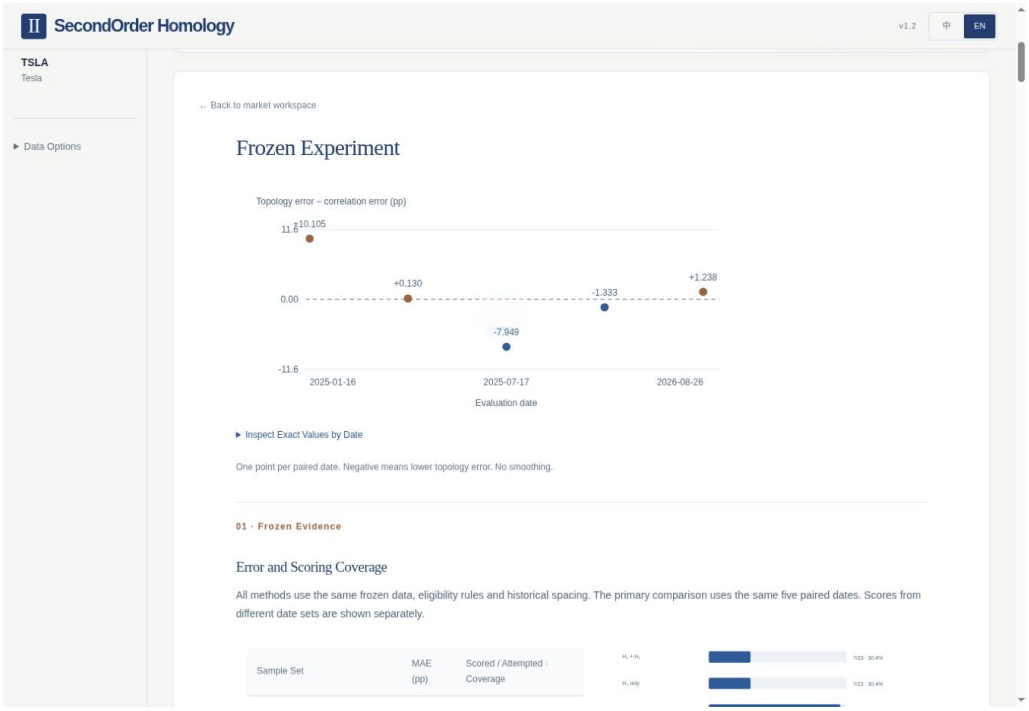


Figure 9: The archived frozen-results page. Points show paired absolute-error differences on the same five scored dates, with negative values favoring topology. This screen displays the historical comparison reported in the paper.

9 Locked prospective evaluation

9.1 Protocol and baskets

A project protocol, `prospective-h1-v1`, was timestamped on October 14, 2026, before its new outcome period. The timestamp belongs to the project archive. Evaluation begins with the first retained aligned date on or after October 14, 2026. The three fixed baskets and prediction targets are listed in Table 5.

Table 5: Fixed prospective baskets. At manuscript preparation on October 22, 2026, the first locked-period 20-observation outcome has not matured. No independent MAE is reported.

Basket	Instruments	Target
Cross-sector equities	JPM, XOM, JNJ, CAT, PG, NEE	JPM
Sector ETFs	XLF, XLI, XLE, XLV, XLP, XLU	XLF
Rates, credit, real assets	IEF, TLT, LQD, HYG, GLD, DBC	TLT

The settings remain $w = 30$, $h = 20$, origin stride 20, candidate stride five, maximum six references, minimum three, channel weights 0.7/0.3, dimension weights 0.5/0.5, eight query resamples, and retention threshold 0.5. Each basket contributes its first 24 mature attempted origins, including abstentions. With this stride and horizon, the complete stage requires approximately 480 retained observations after the first evaluation origin, so a complete result is a long-horizon research commitment.

Inputs must explicitly declare consistently adjusted OHLCV, match the chosen basket, have increasing complete aligned dates, and contain completed observations. Evaluation inputs must have nonsynthetic provenance, with source conventions and exclusions recorded. Previously matured evaluation-period information may enter later historical reference banks under the same past-only rule; model settings remain fixed.

9.2 Endpoint and decision rule

The primary prospective endpoint is $\Delta_{\text{full},0}$ on shared scored dates. Full topology versus labelled geometry is secondary. Each basket is evaluated separately after its first 24 mature attempts. The locked support rule requires at least 20 shared scored origins, upper bounds below zero for all three 95% paired circular-block percentile intervals, and full coverage no more than five percentage points below H_0 coverage. A conclusion of support across the three baskets requires each basket to meet that rule. Otherwise, results are reported as unsupported or inconclusive according to their maturity and paired count, with all denominators.

This criterion is intentionally demanding relative to the observed development coverage. Seven of 23 shared full/H_0 scores in the frozen sample and nine of 32 in the extension are far below 20 of 24. If similar coverage persists, the prospective stage is likely to remain inconclusive under its locked rule. Changes to the gate or stopping rule require a new protocol identity.

The baskets are analyzed separately because they share market conditions and evaluation conventions. The support rule is an operational criterion; correlated intervals and baskets complicate any global error-rate interpretation. Prespecifying the comparison limits further model selection during evaluation [18, 41].

10 Discussion

10.1 Compression, identity, and the prediction target

The unchanged predictions focus attention on what the representation retains and how that information relates to the target.

Proposition 1 identifies a possible mismatch between representation and target. A labelled target such as META may occupy a different structural role across windows whose unlabelled persistence diagrams are identical. A diagram represents aspects of the overall dependence shape, while the forecast endpoint concerns a specific instrument. Whether invariance is useful depends on the relation between structural roles and target outcomes. This connects the structural emphasis of financial network and topology studies [5, 10, 12] to the choice of a target-specific outcome.

The small asset spaces also constrain what can appear. Eight assets produce relatively small Rips complexes, and H_1 is absent in most frozen windows. The finite H_0 diagram reduces a hierarchy to merge values, while H_1 records only loops that survive over a positive threshold interval. Even when present, a short lifetime can be sensitive to correlation estimation and need not correspond to a distinct economic mechanism. Mathematical diagram stability and statistical feature inference are relevant but separate considerations [26, 27, 29].

10.2 Coverage and practical usefulness

Abstention protects the procedure from presenting some unstable comparisons as usable estimates. Its cost is reduced availability. In both historical samples, full topology scores fewer than one third of attempted dates. A useful deployment claim would therefore need an explicit decision problem describing what happens on unscored dates, the cost of abstention, and the value of any returned estimate. They would complement the MAE and retention statistics reported here.

Selective-prediction theory provides the right reporting principle, but our gate is not calibrated to guarantee a target regression risk [14, 15]. With only eight resamples, changes around a threshold of 0.5 are coarse. Query-only perturbation against a fixed shortlist also leaves uncertainty in historical candidate estimation largely unexplored. A separate diagnostic study could assess these sources of uncertainty alongside the acceptance rate.

10.3 Data limitations and inferential scope

Common-date exclusions change the observation clock. Returns across missing sessions and outcomes of different calendar durations complicate the interpretation of a nominal 20-step target. Provider revisions and present-day instrument selection further restrict claims about deployable historical performance. Strict past-only reference chronology addresses explicit outcome leakage, while a point-in-time market study would additionally require appropriately archived data and an investable universe available at each date.

The five- and six-date paired comparisons are too small for a general forecasting ranking or an equivalence claim. Identical full/H_0 predictions describe the archived runs, while shared observations and candidate banks limit effective independence. Design choices made during development also create the model-selection concerns documented in financial backtest research [16, 17, 40, 41]. A trading evaluation would additionally need a specified decision rule, transaction costs, and risk measures.

10.4 Focused further work

A larger forecast study would need a point-in-time input archive, a broader set of targets and market conditions, prespecified alternatives such as an identity-preserving topological representation, and an uninspected evaluation period.

A useful next experiment would control the link between geometry and outcome. It should distinguish a data-generating process in which the target depends on labelled asset roles from one in which the relevant information is invariant under relabelling, and include an outcome-independent control. This would test when topological compression helps prediction, extending the geometric illustration in Figure 2 to a controlled outcome model.

Other methodological comparisons can then be justified by a specific limitation. Covariance regularization and dynamic dependence models address estimation [21, 22]; robust measure-based constructions address other geometric noise models [30]; landscapes and images offer alternative topological summaries [31, 32]. Broader topological learning methods provide further context [33]. These alternatives can be compared on a common, prespecified set of evaluation dates.

11 Conclusion

Persistent loops appear in the archived market geometries, but adding them to the fixed H_0 retrieval procedure leaves every scored prediction unchanged. The frozen comparison has identical ordered references at 23 attempted origins and identical predictions on seven scored dates. The overlapping extension has identical reference sets at 32 attempted origins, one order change on an abstained date, and identical predictions on nine scored dates. These findings distinguish a feature's occurrence from its effect on the returned estimate.

The representation arguments help interpret that distinction. Persistence discards asset labels, H_0 retains merge values, and empty loop diagrams preserve retrieval order by a common rescaling. The empirical records cover cases beyond that empty-loop condition and show what the complete fixed procedure actually returned. The accompanying checkers and derived records expose the numerical fixtures, acceptance denominators, references, and errors underlying these claims.

Combined topology has slightly higher error than labelled geometry on the five and six shared scored dates, with coverage below one third. Small paired samples and overlapping history prevent a general forecasting ranking. These runs document a richer representation that changes no scored prediction. The evidence separates feature occurrence, retrieval, coverage, and outcome error; the locked prospective protocol supplies a distinct future test.

Data and code availability

Verification materials accompany the source submission in `anc/verification/`. They include the two independent Python checkers, archived derived records for all four methods and both samples, retained-date lists, selected persistence fixtures, the prospective protocol, figure-generation scripts, and a script exporting the manuscript's empirical tables and complete returned-reference lists. The README specifies commands, dependencies, expected outputs, source identities, and the limits of each check. The package requires no provider credential or network request. These materials make the reported arithmetic, denominators, returned lists, and fixture comparisons inspectable from the supplied records [44].

The research website and frozen replay are available at https://homology.secondorder.tools/?lang=en. Raw Marketstack OHLCV and the complete acquisition-and-retrieval engine are not redistributed in the supplementary package. Reproduction from raw prices therefore requires additional materials. The supplied data audit documents the exclusion-record gap described in Section 3.2. Dataset fingerprints and calculation identities are given in Appendix D.

Tool assistance

ChatGPT (OpenAI) was used to assist with manuscript drafting and editing, LaTeX preparation, and supporting analysis and verification scripts. The empirical checks and their scope are described in Section 6.3. The author is responsible for the submitted content.

A Representation arguments

A.1 Permutation invariance

For any threshold ϵ, a permutation of vertices maps a simplex of $\mathrm{VR}_\epsilon(D)$ to a simplex of $\mathrm{VR}_\epsilon(\Pi D \Pi^\mathsf{T})$. This is a simplicial isomorphism and is compatible with the inclusions between thresholds. The resulting persistence modules are isomorphic, so their finite diagrams agree. The labelled comparator instead compares each asset pair with the same named pair in the second window. Permuting one window can change those entries. The four-point example in Figure 2 provides a concrete case.

A.2 Finite components and minimum spanning trees

At a given threshold, H_0 records the connected components of the threshold graph. Processing edges in nondecreasing order merges two components exactly when the edge joins distinct components. Kruskal's minimum-spanning-tree procedure accepts precisely such edges. Consequently, component death thresholds agree with the merge-edge weight multiset. Equal weights may allow multiple trees with the same relevant merge thresholds. Edges at zero produce zero-lifetime classes, which

the implementation omits, and one component remains essential. A list of merge weights cannot reconstruct the full labelled tree or all intermediate component memberships.

A.3 Empty loops and retrieval

Under the assumptions of Proposition 3, $u_{L,1} = u_{R,1} = 0$ for every query–candidate comparison. Substituting into Equation (9) gives $D^2_{\text{full}} = D^2_0/2$. Multiplication by the same positive constant preserves order and exact ties. The greedy separation rule therefore sees the same candidate sequence and returns the same references. This conclusion concerns reference selection; acceptance gates that involve newly resampled queries require their own check.

For the analytic square, edges of length $\sqrt{2}$ connect all four vertices into a cycle. Three edges merge components, producing three finite component deaths at $\sqrt{2}$. At that threshold the graph has no triangles, so the cycle survives. At threshold 2, both diagonals and all triangles enter and the cycle is filled, producing the single finite point $(\sqrt{2}, 2)$. This supplies an exact nonempty-H_1 fixture compatible with a valid Gram correlation matrix.

B Parameters and numerical-check scope

Table 6: Fixed historical and prospective retrieval settings. The linear clipping rule is window-local. The H_0-only distance has no dimension factor of one half; this changes scale rather than ranking when H_1 is absent.

Parameter	Value
Retained prices per query	30
Retained-step outcome horizon	20
Evaluation-origin stride	20
Candidate-endpoint stride	5
Maximum / minimum references	6 / 3
Reference-endpoint separation	50
Linear / rank channel weights	0.7 / 0.3
Full-distance H_0 / H_1 weights	0.5 / 0.5
Clipping distance from median	4 robust scales
Query stability resamples / block length	8 / 5
Top-reference / set retention minimum	0.5 / 0.5
Paired loss bootstrap replicates	2,000
Paired bootstrap block lengths	1, 2, 3
Minimum shared origins for intervals	20

The independent checker reconstructs vertices, edges, and triangles from each exported symmetric distance matrix. It reduces boundary sets over $\mathbb{F}_2$ in a separate Python implementation and compares positive finite diagrams to the exported JavaScript results. For each diagram-distance fixture, it builds the diagonal-augmented cost matrix and solves it using SciPy's assignment routine. The 99 matrix fixtures comprise 48 observed channel geometries, 50 generated unit-vector geometries, and one analytic square; the latter generated objects are computational fixtures, not synthetic market outcome experiments.

At absolute tolerance 10^{-12}, the maximum diagram-coordinate discrepancy is zero and the maximum distance discrepancy is $5.551115123125783 \times 10^{-17}$.

C Complete frozen-origin reporting

Table 7 provides the complete frozen attempted-origin status. A score is denoted S, and the primary abstention reasons are U for unstable references and F for too few separated references. Additional failed checks remain in the machine-readable evidence. The reference-set Jaccard similarity between full topology and H_0 is 1.000 at every listed origin; the full ordered lists also agree.

Table 7: All frozen attempted origins. These dates index query endpoints, not outcome endpoints.

Endpoint	Full	H_0 only	H_1 only	Labelled
2024-09-30	U	U	F	F
2024-11-07	F	F	S	F
2024-12-11	S	S	S	U
2025-01-16	S	S	S	S
2025-02-19	U	U	S	U
2025-03-19	U	U	S	U
2025-04-16	S	S	S	S
2025-05-19	U	U	S	S
2025-06-17	U	U	S	U
2025-07-17	S	S	S	S
2025-08-14	S	S	S	S
2025-09-12	U	U	S	S
2025-10-10	U	U	S	U
2025-11-07	S	S	S	U
2025-12-08	U	U	S	U
2026-01-08	U	U	S	U
2026-02-09	U	U	S	U
2026-03-10	U	U	S	S
2026-04-10	U	U	S	S
2026-05-12	U	U	S	S
2026-06-30	U	U	S	S
2026-07-29	U	U	S	U
2026-08-26	S	S	S	S

D Study identities and reproducibility

The frozen study uses the immutable adjusted-data record dated October 13, 2026, calculation release 0.21.0. Its target is META. Its dataset SHA-256 is

```
7263ff324a2c7c38f0c34a87c2cbb7bf
ed906fd49fe2b8eeb76778685423b47b.
```

The extended input uses the later historical acquisition archived on October 14, 2026. Its dataset SHA-256 is

```
86c89d807716ca1e65599c2670b14fe8
29610f84cd9a23055e1821aa1092cffd.
```

Each two-line string is one concatenated 64-character hash; the final period is punctuation. The frozen original is preserved independently of the extension's provider revisions and date-level reconstruction. These fingerprints identify input objects, not the rendered PDF or manuscript source.

The supplementary directory `anc/verification/` contains a README and requirements file. From that directory, run:

```
python verify_archived_results.py
python verify_numerics.py
python export_results.py
```

The first command checks scored counts, errors, returned-list comparisons, and reference chronology. The second checks exported filtration and assignment fixtures. The third rebuilds the empirical data tables and exports the rows underlying each comparison. Optional figure-generation commands are documented in the README. Their outputs are separate from the archived input records. The commands use the supplied local inputs.

A manuscript build uses `latexmk -pdf market-matching.tex`, or pdfLaTeX, BibTeX, and pdfLaTeX twice. The supplementary scripts run separately from the TeX build.

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